# Muhammad Nabeel Malik

Full-Stack AI Engineer | LLM Agents & Full Stack Development

mnmalikdev@gmail.com +32 469 12 73 83 Bayreuth, Germany

LinkedIn | GitHub | Portfolio

## Professional Summary

Full-stack software engineer with 5+ years building SaaS, fintech and e-commerce products, now focused on AI engineering: LLM agents, RAG and tool-calling systems in production. Comfortable across the stack, from scalable backends and data-heavy PostgreSQL to modern frontends and cloud deployments, with React.js, Next.js, Nest.js, Python, Kafka, AWS and Docker.

## Work Experience

**Full-Stack AI Engineer (full-time),** *OSPHIM* 07/2026 – Present
Aachen, Germany

- Built an LLM agent (Python, FastAPI, SQLAlchemy, Claude) with tool calling and tenant-scoped data access that explains production data and flags gaps instead of guessing.
- Built a voice pipeline turning spoken shift handovers into structured data with speech-to-text and LLM extraction.
- Moved 6 product areas from a 1.4B-row raw table to roll-up tables, cutting dashboard load from 1,150 ms to 16 ms and filter time from 2,240 ms to 14 ms.
- Fixed set-point detection that missed changes on long jobs by scanning only the first 5,000 readings.

**Full-Stack Engineer (part-time),** *OSPHIM* 01/2026 – 07/2026
Aachen, Germany

- Rebuilt the production logbook so one entry links 5+ entity types, on a redesigned PostgreSQL schema (Svelte, NestJS).
- Migrated logbook content from plain text to JSONB with zero downtime and 100% data integrity (Drizzle ORM).
- Built a digital shift handover protocol, replacing paper notes with structured handovers.
- Extended experiment setup from single-cavity to multi-cavity molds: one setup for 16 cavities instead of 16.
- Tuned PostgreSQL for 240M+ monthly time-series records with partitioning and vacuum tuning.

**Software Engineer,** *MobileLive* 05/2024 – 10/2025
MonieCorp (Fintech Product) Remote

- Developed responsive and modular UI components in React + TypeScript + ShadCN, improving page load speed by 25% and reducing UI bugs by 40% after refactoring.
- Built 15+ SOLID-compliant React components for Admin panel using Cursor IDE for rapid development, cutting client feedback cycles from 3 days to 24 hours.
- Built multi-step React form for admins to configure tiered currency margins with validation; streamlined setup from 20+ minutes to <4 minutes per currency pair.

Staples Canada (MobileLive Client)

- Implemented Mixpanel tracking for 100,000+ users, capturing product views, add-to-cart, and checkout events, enabling business team to optimize flows and boost conversions by 12%.
- Built custom React features using Canva API for Staples Online Print Shop, enabling dynamic product personalization and increasing user engagement by 20%.
- Fixed 30+ production bugs within 4-hour SLAs, maintaining 99.9% uptime during peak sales.

**Associate Software Engineer,** *Code-Huddle* 08/2022 – 05/2024
GYMYG: Virtual Gym Partner (US-Based Startup) Islamabad, Pakistan

- Built Nest.js APIs with JWT RBAC, handling 10K DAU at <200ms P95 latency.
- Integrated Stripe monthly subscriptions, processing $15k+ in secure payments.
- Reduced video load times by approximately 30% using FFMPEG compression and optimized AWS CloudFront edge delivery.
- Automated end-to-end CI/CD pipeline with Github Actions, AWS CodePipeline, cutting deployment time by 40%.

Ordinami: QR Ordering Web-App (Italian Startup MVP)

- Implemented Firebase Auth + Firestore backend, supporting 100+ daily orders.
- Built an admin-panel shadcn/ui + Tailwind for menu creation, reducing manual setup time for owners by 60%.

**Product Development Intern,** *Code-Huddle* 04/2022 – 06/2022
Islamabad, Pakistan

- Assisted in frontend (React.js) and backend (NestJS) development, contributing to the design and implementation of scalable solutions.
- Supported AWS operations and collaborated on ongoing projects, gaining exposure to the full product development lifecycle.

## Education

**Bachelor of Science in Software Engineering,** *COMSATS University* 2018 – 2022
Islamabad, Pakistan

## Skills and Core Competencies

**AI & LLM Engineering**
Python | FastAPI | SQLAlchemy | LLM Agents | Agent Harness | Tool Calling | RAG | LLM Extraction | Speech-to-Text

**Frontend Development**
React.js | Next.js | SvelteKit | Context API | Tailwind CSS | HTML5 | CSS3 | Shopify (Liquid Templating) | ShadCN UI | Redux and Redux Toolkit | RxJs

**Software Development & Engineering**
Full-Stack Web Applications | RESTful APIs | Microservices Architecture | Secure Authentication Systems | Payment Integrations (Stripe) | CI/CD Automation | SaaS Product Engineering

**Cloud, DevOps & Containerization**
AWS (ECS, ECR, EC2, RDS, S3, CloudFront, CodePipeline, CloudWatch, SNS) | Azure | Docker

**Backend Development**
JavaScript | TypeScript | Node.js | Express.js | Nest.js | TypeORM | Prisma | Drizzle ORM | Kafka | Redis

**Databases & Data Management**
MySQL | PostgreSQL | MongoDB | Query Optimization | Data Modeling

## Notable Freelance Projects

**Pendfunds.com (M.V.P.),** *Contract Management & Mediation Software* 01/2024 – 09/2024

- Developed full escrow payment flow with Stripe Connect and Optimus Bank merchant payouts in Nest.js, onboarding 100+ users and handling hundreds of transactions.
- Automated deployments across development, staging, and production environments using AWS Elastic Beanstalk and AWS CI/CD pipelines.
- Implemented secure payload encryption/decryption using PGP with RSA-256, ensuring sensitive transaction data remained fully protected.